

ACTIVITY 6 — FACILITATOR GUIDE

Data Collection, KPIs, Check Sheets, and Basic Metrics

TYPE	DURATION	DMAIC PHASE	MAPS TO
Core Lab	45 min	MEASURE	A4

Purpose of this activity

Plan and execute data collection against defined quality standards (A4).

Replace anecdote with evidence. Define KPIs with unambiguous operational definitions, design a check sheet, plan the sampling approach, and understand which data type you are collecting — because the data type drives the tool.

What learners produce

A data collection plan with operational definitions, a working check sheet and a sampling plan with the bias risk named.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- A laptop with Excel, Google Sheets or LibreOffice Calc
- The CSV files in this activity's data/ and templates/ folders
- Flipchart or A3 sheet and markers
- The course slides for the scenario and the framework

Set-up

Groups of 3-5. Distribute one worksheet per learner and one checklist per group. Every activity runs on the same continuous Contoso Service Desk scenario, so remind learners where this one sits in that story before starting.

Scenario / briefing material

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder

data/ — the mock data you analyse:

- daily-summary.csv — 10 rows (date, tickets, mean_assignment_min, median_assignment_min, defects, missed_30min_goal)
- service-desk-tickets.csv — 400 rows (ticket_id, date, ticket_type, channel, assigned_queue, agent, assignment_time_min, met_30min_goal, defect_count, defect_categories, reopened)

templates/ — the worksheets you complete:

- check-sheet.csv (date, ticket_id, ticket_type, assignment_time_min, defect_category, rework_required, notes)
- data-collection-plan.csv (kpi, operational_definition, data_type, data_source, frequency, who_collects, bias_risk)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

data/service-desk-tickets.csv is the full two-week extract (400 tickets).

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Classify your data: continuous vs discrete, nominal vs ordinal — and why it matters.	The operational-definition test is the whole point: two people, same event, same value.
2	Define KPIs with operational definitions, data source and collection frequency.	Overlapping defect categories are the usual failure — test with a borderline example.
3	Design a check sheet capturing date, ticket ID, type, assignment time and defect category.	
4	Define mutually exclusive defect categories so every observation lands in exactly one.	
5	Plan sampling: how many, how often, by whom — and identify possible bias.	
6	State which KPI best reflects the customer pain point from your VOC work.	

Suggested timing

Minutes	Phase
8 min	Classify the data types
12 min	Define KPIs with operational definitions
10 min	Design the check sheet
8 min	Define mutually exclusive defect categories
7 min	Plan the sampling and name the bias risk

Total: 45 minutes

Discussion & decision prompts

1. Would two different people record the same value from your operational definition?
2. Which KPI best reflects the customer pain from your VOC work?
3. Where could bias enter your sampling, and who would introduce it?

Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what the data showed.
- Link the activity explicitly back to the MEASURE phase of DMAIC.
- Remind learners this activity rehearses what the Case Study assessment will require.

Reflection questions:

- What is measured at your workplace that nobody has operationally defined?
- Who collects that data, and what incentive do they have to shade it?

Facilitator notes — what to watch for

- The operational-definition test is the whole point: two people, same event, same value.
- Overlapping defect categories are the usual failure — test with a borderline example.

Success criterion

Two different people reading your operational definition would record the same value for the same event.

Assessment link

This activity builds the MEASURE phase of DMAIC. All activities are in-class activities assessed indirectly through the open-book Case Study (CS), which reuses the same techniques on the same Contoso Service Desk scenario and the same figures (400 tickets, yield 76%, DPMO 50,000). Knowledge is assessed in the Written Assessment (WA-SAQ). Nothing is assessed that is not practised here or taught in the slides.